RO Membrane Fouling Early-Warning Model

Technical report | Prototype v0.1.0 | 28 October 2026

1. Objective and scope

This project packages a Python command-line tool and REST API for time-stamped reverse-osmosis (RO) data. The synthetic-data prototype calculates a simplified normalized permeate-flow indicator, predicts seven-day flow behavior, reports a separate 14-day linear trend projection, and estimates time to a configurable threshold. A second, independent workflow analyses real-station RO production flow. Neither workflow is validated for operational maintenance decisions.

2. Data assessment

Public candidate sources reviewed include the U.S. Department of Energy/NREL Experimental Wave Desalination dataset and Queen's University Belfast SWRO performance workbooks. Their landing pages do not establish a long-term, labelled fouling/cleaning time series. An industrial fouling study reports SCADA analysis with data available on request. A 422-day ultrafiltration dataset was not used to train an RO model because UF and RO are different processes. Downloadable public files have not been fully inspected, so none is claimed as the training source for this prototype.

The synthetic workflow uses artificial daily data with varying temperature, pressure and feed TDS, gradual degradation, noise and partial cleaning resets. These assumptions make the software pipeline testable but do not reproduce all physical fouling mechanisms.

3. Normalization and synthetic-model validation

The prototype estimates net driving pressure (NDP) from feed/concentrate pressure, permeate pressure and an approximate osmotic-pressure term proportional to TDS. It applies a simplified temperature correction referenced to 25 °C and an NDP reference of 20 bar. The first-order TDS coefficient was corrected from a factor-of-1000 configuration error; 35 g/L now corresponds to approximately 26.25 bar. This is a rough approximation, not a verified ASTM D4516 or membrane-vendor method.

For the synthetic experiment, 286 supervised rows were split chronologically into 200 training, 43 validation and 43 test rows. Random Forest validation MAE was 0.02077 normalized-flow ratio, versus 0.00794 for persistence. Persistence was therefore selected using validation data only. On the held-out synthetic test segment, selected model MAE was 0.00822 and RMSE was 0.00975. These metrics describe the synthetic generator only.

Synthetic model selection is not evidence of real-plant forecasting accuracy or fouling-detection performance.

Real-Station Results, Limitations and Packaging

4. Real-station production-flow experiment

The separate station dataset contains 540 daily observations from 4 January 2024 through 26 June 2025, with no missing calendar dates recorded and 19 days with zero RO production. It contains production flow and operational phase information, but no confirmed fouling labels or verified cleaning-event labels. It is therefore used for production-flow forecasting and rule-based low-flow screening, not as ground truth for membrane fouling.

Recorded result	Value
Chronological split	364 train / 71 validation / 71 test
Validation MAE — Random Forest	529.36 m³/day
Validation MAE — persistence	258.24 m³/day
Selected model	Persistence baseline
Test MAE — selected baseline	97.02 m³/day
Test RMSE — selected baseline	183.42 m³/day

The model was selected using validation MAE, not test results. Although Random Forest had a lower recorded test RMSE, that result does not justify changing the selection after examining the test set. Metrics apply only to this dataset and split; generalization to another plant has not been established.

5. Rule-based alerts and failure modes

The recorded low-flow analysis produced 47 alert days, 47 watch-low-flow days and 19 production-stopped days. These are rule outputs, not verified fouling events. The rolling baseline may drift during sustained low production. Flow changes can also result from cleaning, feedwater changes, sensor drift, hydraulic changes, pump faults or other interventions. Trend extrapolation can fail after abrupt changes.

6. Software packaging and reproduction

The repository includes a Python package, CLI, FastAPI API, Dockerfile, YAML configuration, versioned joblib artefacts, sample CSV, data-source log and automated tests. The API exposes GET /health and POST /predict; the separate real-station analysis is available through the analyze-real CLI workflow. Input validation checks required columns, timestamps, duplicates, numeric values, plausible ranges, positive NDP and excessive time gaps.

Reproduction from the project directory: python -m pip install -e '[dev]'; python -m pytest -q; fouling predict data/sample_input.csv. For the API: uvicorn fouling.api:app --host 127.0.0.1 --port 8000. See README.md for real-station and Docker commands.

7. Limitations and next steps

Neither workflow independently diagnoses membrane fouling or determines chemical-cleaning requirements. Before operational use, inspect source datasets at row/sheet level, obtain labelled real-plant fouling and cleaning events, validate across unseen time windows and plants, replace simplified normalization with an appropriate membrane/vendor method, quantify threshold-crossing uncertainty, and obtain process-engineering review.

Sources reviewed: NREL/Data.gov experimental wave desalination data; Queen's University Belfast SWRO performance work; a 2026 industrial RO fouling study; and a Dryad UF pretreatment dataset. Source qualifications and access notes are documented in data/data_source_log.md.